



A comprehensive review of sustainable bioremediation techniques: Eco friendly solutions for waste and pollution management

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ABSTRACT

Bioremediation, an advanced and environmentally sustainable technology, utilizes biological microorganisms to mitigate pollution. This review combines insights from two perspectives: one focusing on the mechanisms, applications, and types of bioremediation, and the other examining the transformative potential of integrating Internet of Things (IoT), Artificial Intelligence (AI), and biosensors in pollution management. The first perspective delves into the effectiveness of bioremediation in decomposing and detoxifying hazardous substances, emphasizing its cost-effectiveness and eco-friendliness compared to conventional methods. In-situ and ex-situ bioremediation methods are analyzed, along with intrinsic and engineered techniques, and phytoremediation strategies for heavy metal removal. The review underscores the growing importance of bioremediation in addressing industrial effluents, contaminated soils, and groundwater, with future advancements expected to enhance its efficiency and applicability. From the second perspective, recent advancements in IoT, AI, and biosensors are explored for their potential to revolutionize bioremediation and waste management. IoT facilitates real-time monitoring and remote management, AI enhances data analysis and predictive modelling, and biosensors contribute to precise pollutant detection and environmental monitoring. The review highlights the synergistic integration of these technologies, presenting smart bioremediation systems with real-time feedback loops and adaptive capabilities. Together, these technologies offer scalable solutions for environmental pollution mitigation, marking a significant stride towards sustainable environmental management.

Introduction

Bioremediation represents a cutting-edge and ecologically sustainable technology that utilizes biological microorganisms to decompose and neutralize pollutants (Ajona and Vasanthi, 2021). This technique plays a crucial role in removing environmental toxins, thereby restoring ecosystems to their natural state and preventing further contamination (Sardrood et al., 2013; Arregui et al., 2019; Berillo et al., 2021). The application of bioremediation has expanded rapidly, addressing a wide range of issues such as acidic mine drainage, leaks from underground pipes, and contamination from crime scenes. Through the metabolic activities of microorganisms and their enzymes, hazardous substances are primarily broken down via oxidation–reduction reactions.

Recent advancements in the field have underscored the effectiveness

of bioremediation in mitigating the impacts of anthropogenic pollutants resulting from industrial and agricultural processes (Raajasubramanian et al., 2015). Compared to traditional chemical and physical remediation methods, bioremediation is more eco-friendly, easier to implement, and less invasive. The processes involved in bioremediation are capable of eliminating, detoxifying, and reducing the toxicity of harmful substances, including industrial effluents, heavy metals, xenobiotic compounds, organic halogens, pesticides, and chlorinated compounds (Sharma, 2020).

Innovative research has broadened the scope of bioremediation by exploring the use of genetically engineered microorganisms and consortia of microbes to enhance degradation efficiency and target a wider range of contaminants. Recent publications have highlighted the potential of synthetic biology in creating microbial strains with optimized

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pathways for pollutant degradation, which could significantly advance the field. Additionally, the integration of bioremediation with phytoremediation using plants to assist microbial activity has shown promise in tackling complex contamination scenarios, offering a synergistic approach to environmental cleanup.

Further studies have explored the role of bioinformatics and machine learning in predicting and optimizing bioremediation processes. By analyzing large datasets on microbial genomes and environmental conditions, researchers can identify key factors that influence bioremediation efficacy and tailor interventions more precisely. This data-driven approach represents a significant step forward in enhancing the efficiency and cost-effectiveness of bioremediation.

Bioremediation continues to evolve with technological advancements, expanding its applicability and efficiency in restoring contaminated environments. It remains a crucial strategy for addressing the diverse challenges posed by environmental pollution, leveraging the power of microorganisms to clean up and protect our ecosystems. However, several challenges and gaps persist in its application, hindering its widespread adoption and efficacy.

1. **Complexity of Contaminants:** Many contaminants in polluted sites are complex mixtures of chemicals, making them difficult to degrade completely using bioremediation techniques (Priya et al., 2024). Some contaminants, such as heavy metals and persistent organic pollutants, may require specialized microbial consortia or genetically engineered organisms for effective degradation (Nag et al., 2024).
2. **Site Heterogeneity:** Environmental conditions at contaminated sites can vary greatly, with differences in pH, temperature, moisture content, and nutrient availability. Developing bioremediation strategies that can adapt to these variations and maintain consistent performance is a considerable challenge (Ceballos-Escalera et al., 2024).
3. **Slow Degradation Rates:** Bioremediation processes can be slow, particularly for recalcitrant contaminants (Geed et al., 2024). Microorganisms may require time to adapt to the presence of contaminants and establish populations capable of efficient degradation. Additionally, factors such as limited nutrient availability or toxic conditions can further slow down degradation rates.
4. **Limited Applicability to Certain Contaminants:** While bioremediation is effective for many organic contaminants, its applicability to certain types of pollutants, such as heavy metals and radioactive elements, is limited. These contaminants may require alternative treatment approaches, and integrating bioremediation with other remediation techniques may be necessary for comprehensive cleanup (Milić et al., 2024).
5. **Risk of Secondary Contamination:** In some cases, bioremediation processes can result in the release of partially degraded intermediates or transformation products, which may be more toxic or mobile than the original contaminants (Milić et al., 2024). Understanding and mitigating the risk of secondary contamination is essential to ensure that bioremediation efforts do not inadvertently worsen environmental conditions (Liu, et al., 2024).
6. **Regulatory Challenges:** Regulatory frameworks governing the use of bioremediation techniques vary between regions and may present barriers to implementation. Ensuring compliance with regulatory requirements, obtaining necessary permits, and addressing liability concerns can add complexity and cost to bioremediation projects (Prasad, 2024; Tedesco et al., 2024).
7. **Monitoring and Verification:** Assessing the effectiveness of bioremediation treatments and ensuring that cleanup goals are achieved require robust monitoring and verification protocols (de Souza et al., 2024). However, existing methods for monitoring microbial activity and contaminant degradation in situ may be limited in their accuracy, sensitivity, or ability to provide real-time data.

To address these challenges and fill existing gaps in bioremediation technology, ongoing research efforts should focus on several fronts

1. **Microbial Ecology and Genomics:** Advancements in microbial ecology and genomics are enhancing scientists' understanding of the diversity and functions of microbial communities involved in bioremediation processes (Xue et al., 2024). This knowledge can inform the selection of microbial consortia or the development of genetically engineered organisms tailored to specific contaminants and environmental conditions (Patel Desai., & Thakur, 2024).
2. **Bioinformatics and Computational Modeling:** Bioinformatics tools and computational models are being used to predict the behaviour of microbial populations in response to environmental conditions and contaminant exposure and these techniques can be explored to the maximum (Pazos et al., 2024). Bioinformatics, IOT, Biosensors, learning tools can aid in the design of optimized bioremediation strategies and the prediction of long-term treatment outcomes (Oyewusi et al., 2024) According to Srinivasan, et al., 2024 nanobiosensors can be used to detect pesticides in the field and this technology can also be suitably modified, explored and exploited in bioremediation.
3. **Bioreactor Design and Engineering:** Advancements in bioreactor design and engineering are facilitating the creation of controlled environments where bioremediation processes can be optimized and scaled up (Sandoval-Herazo et al., 2024). Innovative bioreactor configurations, such as bioelectrochemical systems and membrane bioreactors, present opportunities to enhance contaminant removal efficiency and reduce treatment times (Orzechowska et al., 2024). Developing the most efficient bioreactor requires integrated advanced research from scientists with diverse expertise, utilizing a transdisciplinary approach.
4. **Bio stimulation and Bioaugmentation:** Biostimulation techniques, such as the addition of nutrients or electron acceptors, can enhance microbial activity and accelerate contaminant degradation rates in bioremediation systems. Similarly, bioaugmentation involves introducing exogenous microorganisms with specific metabolic capabilities to boost biodegradation processes in contaminated environments (Omenna et al., 2024).
5. **Advanced Monitoring Technologies:** Research into advanced monitoring technologies, such as molecular biomarkers, stable isotope probing, and remote sensing techniques, is enhancing our ability to track microbial dynamics and contaminant transformation processes in real time. The integration of Artificial Intelligence (AI) and Internet of Things (IoT) devices offers potential for experimentation, with the goal of making these technologies more cost-effective and scalable (Ukhurebor et al., 2021). These innovations provide valuable insights into the effectiveness of bioremediation treatments and support adaptive management approaches, not only in bioremediation but also in the design and optimization of bioreactors (Kumar et al., 2024).

By addressing these challenges and leveraging emerging technologies, bioremediation holds significant promise as a cost-effective and environmentally sustainable approach for addressing a wide range of contaminated sites. However, continued interdisciplinary, trans disciplinary integrative, collaboration and innovation will be essential to overcome existing, upcoming barriers and realize the full potential of bioremediation in environmental remediation efforts.

Bioremediation, bolstered by the integration of cutting-edge technologies, offers promising solutions to address critical environmental challenges (Agrawal et al., 2024). By harnessing advanced biotechnological tools such as genomic analysis, artificial intelligence (AI), and precision monitoring systems, bioremediation efforts can achieve more precise and efficient degradation of pollutants in air, water, and soil. This integrated approach allows for customized strategies that not only

improve remediation effectiveness but also optimize resource utilization, reduce costs, and minimize environmental impact. Furthermore, by merging scientific innovation with practical application, bioremediation emerges as a sustainable strategy to restore ecosystems, protect public health, and mitigate global climate change impacts and also to address the pressing urgency for bioremediation in present environmental context Table 1. (Mohan et al., 2024).

Postulates of bioremediation

Bioremediation is a sophisticated and ecologically sustainable approach that employs living organisms, including plants and microorganisms such as fungi, bacteria, and algae, to degrade and detoxify hazardous chemicals in the environment. These organisms convert pollutants into less harmful substances such as carbon dioxide, water, microbial biomass, and metabolites. The microorganisms involved in bioremediation leverage these pollutants as part of their metabolic processes, ultimately using them for growth and energy production.

The effectiveness of bioremediation hinges on several critical factors

Types of Contaminants: The biodegradation process is significantly affected by the nature of the contaminants, including their chemical structure and toxicity. Organic compounds such as hydrocarbons, pesticides, and solvents are generally more susceptible to microbial degradation compared to inorganic pollutants like heavy metals, which often require specific conditions or additional treatment steps (Haines and Harris, 1987; Brusseau and Artiola, 2019; Azad, Sheikh, and Hai, 2024).

Solubility and Bioavailability: The solubility of pollutants and their bioavailability to microorganisms are crucial determinants of the bioremediation process. Hydrophobic compounds, for example, tend to partition into soil or sediments, making them less accessible to microbes. Enhancing the bioavailability of such pollutants through surfactants or solvents can improve biodegradation rates.

Environmental Conditions: The surrounding environment plays a pivotal role in bioremediation efficacy. Factors such as temperature, pH, oxygen availability, and nutrient levels must be optimized to support microbial activity. For instance, aerobic bacteria require sufficient oxygen levels to efficiently degrade organic pollutants, whereas anaerobic conditions might be necessary for the breakdown of specific compounds like chlorinated solvents.

Microbial Community Dynamics: The composition and diversity of the microbial community at a contaminated site can impact the degradation process. Indigenous microorganisms that have adapted to the presence of pollutants often exhibit higher degradation capabilities. However, bioaugmentation—introducing specialized microbial strains—can enhance bioremediation when native populations are

Table 1
Key data highlight the pressing urgency and significance of bioremediation in present environmental context.

Aspect	Statistic/Data	Source
Air Pollution Levels	Air pollution contributes to 7 million premature deaths annually worldwide.	WHO (World Health Organisation)
Water Pollution	Water pollution affects over 80 % of water bodies globally.	EPA (United States Environmental Protection Agency)
Economic Costs	The cost of environmental degradation is estimated at over \$4.6 trillion annually, about 6.2 % of global GDP.	World Bank
Climate Change Impact	Methane is 84 times more potent than carbon dioxide in the short term, exacerbating climate change.	European Union & IPCC (Intergovernmental Panel on Climate Change)

insufficient.

Regulation of Environmental Variables: To achieve effective and rapid biodegradation, environmental variables must be carefully regulated. This includes controlling moisture levels, aeration, and nutrient supplementation. For instance, adding nitrogen and phosphorus can stimulate microbial growth and activity, thereby accelerating the degradation process (Tyagi and Kumar, 2021).

Recent studies have expanded our understanding of bioremediation mechanisms. Advanced molecular techniques, such as metagenomics and proteomics, provide insights into the functional genes and metabolic pathways involved in pollutant degradation. Additionally, the use of synthetic biology to engineer microbial strains with enhanced degradation capabilities is a promising area of research. These engineered microbes can be tailored to express specific enzymes that break down recalcitrant pollutants more efficiently.

Furthermore, the integration of bioremediation with other remediation technologies, such as phytoremediation and nanoremediation, offers a multifaceted approach to addressing complex contamination scenarios. Phytoremediation, for instance, utilizes plants to extract, stabilize, and transform pollutants, often in synergy with microbial processes. Nanoremediation employs nanoparticles to enhance the availability and reactivity of pollutants, thus facilitating their microbial degradation.

Bioremediation is a dynamic and evolving field that utilizes the natural abilities of living organisms to clean up contaminated environments (Nazir et al., 2024). By understanding and optimizing the factors that affect microbial degradation, we can improve the efficiency and applicability of bioremediation strategies, ultimately aiding in the restoration and protection of ecosystems (Kalogiouri et al., 2024).

The purpose of bioremediation

Bioremediation is process of providing unstinting remedy using a living organism to create healthy environment suitable for all biotic community to live in concord by reducing the contaminants to undetectable, non-toxic, or tolerable levels within regulatory limits or, ideally, to mineralize organic pollutants into its elemental forms.

Key objectives of bioremediation

Recycling of waste: Bioremediation is used in recycling of waste water, industrial effluents, treatment of contaminated soil.

Detoxification of Pollutants: Bioremediation employs microorganisms and plants to convert harmful substances into less toxic or non-toxic compounds. This detoxification process mitigates the environmental and health hazards posed by contaminated sites.

Mineralization of Organic Pollutants: The ultimate goal of bioremediation is the complete mineralization of organic pollutants. This involves breaking down complex organic molecules into elemental forms such as carbon, hydrogen, and several other elements along with carbon dioxide, water, and microbial biomass, thereby eliminating pollutants from the environment entirely.

Restoration of Ecosystems: By removing contaminants that disrupt ecological balance, bioremediation helps restore natural ecosystems. This process re-establishes the conditions necessary for the survival and growth of various organisms, promoting biodiversity and ecosystem health.

Sustainable and Eco-Friendly Approach: Bioremediation is inherently sustainable and environmentally friendly, as it leverages natural biological processes. It reduces the need for harmful chemicals and minimizes environmental disruption compared to traditional chemical and physical remediation methods.

Compliance with Regulatory Standards: Bioremediation ensures that contaminant levels are reduced to meet regulatory limits, safeguarding human health and the environment. Achieving regulatory compliance is essential for the safe reuse of remediated sites for

residential, agricultural, or industrial purposes.

Cost-Effectiveness: Bioremediation often proves to be more cost-effective than conventional remediation methods. It typically requires fewer resources and less energy, making it an economically viable option for large-scale environmental cleanup projects.

Recent advancements in bioremediation

Recent research has brought forth several innovative approaches to enhance the efficacy of bioremediation:

Genetically Engineered Microorganisms: Scientists have engineered microbial strains with enhanced abilities to degrade specific pollutants through genetic engineering. These advanced strains improve the efficiency and expand the scope of bioremediation processes (de Lorenzo, 2022; Adetunji et al., 2024).

Microbial Consortia: Employing diverse microbial communities that work synergistically can enhance the degradation of complex pollutant mixtures. These consortia are capable of addressing a wider range of contaminants and adapting to different environmental conditions (Guo, Ren et al., 2024; Dianatdar and Etemadifar, 2024).

Phytoremediation Integration: Combining bioremediation with phytoremediation harnesses the strengths of both approaches. Plants help with pollutant uptake and stabilization while creating favorable conditions for microbial degradation in the rhizosphere (Khan et al., 2024; Prasad, 2024).

Nanotechnology: Nanoparticles can increase the bioavailability and reactivity of pollutants, aiding their breakdown by microorganisms. Moreover, nanoparticles can act as carriers for microbial cells, enhancing their distribution and activity in contaminated sites (Thulasingh et al., 2023; Ningombam et al., 2024).

Bioinformatics and Machine Learning: Advanced computational tools are employed to predict and optimize bioremediation processes. By analyzing extensive datasets on microbial functions and environmental conditions, researchers can design more effective bioremediation strategies tailored to specific sites (Yu, Boob et al., 2023).

In recent years, significant advancements in bioremediation techniques have emerged, addressing many of the limitations previously encountered.

Here's an overview of some of the latest advancements and how the challenges can be tackled:

Microbial Community Engineering: Researchers are moving away from single microbial strains and increasingly towards engineering complex microbial communities tailored to specific contaminants and environmental conditions (Bustamante-Torres et al., 2023). This approach, termed microbial community engineering, leverages synergistic interactions between diverse microorganisms to enhance degradation capabilities and resilience to environmental fluctuations. By designing microbial consortia with complementary metabolic pathways, researchers can aim to achieve more efficient and robust bioremediation of complex contaminant mixtures (Adetunji et al., 2024).

Genome Editing and Synthetic Biology: Recent advancements in genome editing technologies, exemplified by CRISPR-Cas/Cas9-mediated functional gene editing (Sahoo et al., 2022), have transformed synthetic biology. These innovations enable precise manipulation of microbial genomes to enhance bioremediation capabilities. Researchers can now engineer microorganisms with customized metabolic pathways optimized for degrading specific contaminants or adapting to varying environmental conditions. By incorporating genetic modifications that boost enzyme activity, substrate specificity, or stress tolerance, engineered microbes can markedly enhance bioremediation efficiency and broaden the spectrum of treatable contaminants (Verde-zoto-Prado et al., 2024).

Omics Technologies: Advanced omics technologies such as metagenomics, metatranscriptomics, and metaproteomics offer unparalleled insights into the structure, function, and dynamics of microbial communities engaged in bioremediation processes (Mohanty, et al., 2024).

By scrutinizing the genetic, transcriptional, and protein-level responses of microbial consortia to environmental cues, researchers can pinpoint crucial genes, pathways, and regulatory mechanisms involved in contaminant degradation. This understanding facilitates more informed development of bioremediation strategies, refinement of treatment conditions, and prediction of ecosystem responses to environmental changes (Singh et al., 2020).

Bioelectrochemical Systems (BES): Bioelectrochemical systems (BES) integrate microbial metabolism with electrochemical processes to improve efficiency in contaminant removal and energy recovery. Within BES, microorganisms catalyze redox reactions on electrode surfaces, generating electrical currents that drive contaminant degradation or facilitate electron transfer to external acceptors (Ghangrekar & Neethu., 2020). These technologies offer numerous advantages for bioremediation, including heightened microbial activity, direct electron transfer pathways, and the capability to selectively control microbial community composition and metabolic pathways. Furthermore, BES can be coupled with renewable energy sources like solar or wind power to achieve energy-neutral or even energy-positive bioremediation processes (Baek & Lee, 2024).

Nanobiotechnology: Nanobiotechnology involves the application of nanomaterials, such as nanoparticles and nanocomposites, to enhance the efficiency and specificity of bioremediation processes (Thulasingh et al., 2023). Nanomaterials can serve as carriers for delivering nutrients, electron acceptors, or immobilizing microbial cells in contaminated environments, thereby improving mass transfer rates and microbial activity. Additionally, engineered nanoparticles can catalyze or facilitate chemical reactions involved in contaminant degradation, acting as nanozymes or photocatalysts under specific environmental conditions. However, the potential environmental impacts and risks associated with the use of nanomaterials in bioremediation require careful consideration and mitigation strategies (Vázquez-Núñez et al., 2020).

Overall, these advancements in bioremediation technology hold great promise for addressing the challenges posed by complex contaminants, heterogeneous environments, slow degradation rates, and regulatory constraints. By harnessing the power of microbial communities, synthetic biology, omics technologies, electrochemical processes, and nanobiotechnology, researchers are paving the way for more efficient, cost-effective, and sustainable solutions to environmental pollution. Continued interdisciplinary, transdisciplinary collaboration and innovation will be essential for translating these advancements into practical applications and achieving meaningful impacts on global environmental health.

Recent advances in bioremediation and implementation in live scenario

Oil spill cleanup with Microbes

Following the Deepwater Horizon oil spill in the Gulf of Mexico in 2010, extensive research has focused on harnessing naturally occurring microbes for the bioremediation of oil-contaminated environments. Studies have demonstrated that certain bacteria possess the capability to efficiently degrade hydrocarbons found in crude oil, significantly aiding in the cleanup efforts (Zapata-Peñasco et al., 2023).

For instance, Zapata-Peñasco and colleagues showcased successful applications of microbial bioremediation techniques in treating oil-contaminated soil and water, highlighting the effectiveness of microbial communities in breaking down complex hydrocarbons (Zapata-Peñasco et al., 2023). Concurrently, Rezaei Somee et al. (2021) conducted genome-resolved comparative analyses of water and sediment samples across the Persian Gulf's pollution gradient. Their findings elucidated the genetic diversity and metabolic potential of microbial communities involved in mitigating oil pollution, underscoring the role of microbial genomics in understanding and optimizing bioremediation

strategies (Rezaei Somee et al., 2021). These studies collectively underscore the promising role of microbial bioremediation in addressing oil spills and emphasize the importance of ongoing research to refine techniques and enhance environmental recovery efforts worldwide.

Heavy Metal Remediation using Plants: Phytoremediation, a bioremediation method employing plants to extract contaminants from soil and water, has garnered attention for its effectiveness in addressing heavy metal pollution. Yang et al. (2022) reviewed the utilization of plants such as sunflowers, willows, and Indian mustard for absorbing heavy metals such as lead, arsenic, and cadmium from polluted sites. Shakoor et al. (2017) and Jensen et al. (2009) documented the successful application of phytoremediation techniques using willow trees to clean up heavy metal-contaminated soil in industrial areas. Additionally, Jiang (2024) contributed research highlighting similar successful applications of willow trees in this context.

Biodegradation of Pesticides: Contamination remains a critical environmental issue in agricultural regions. Bioremediation techniques leveraging microbial degradation have emerged as promising approaches to mitigate pesticide pollution (Kumar Goutam et al., 2024). Recent research has concentrated on identifying and enhancing microbial strains capable of degrading specific pesticides like atrazine and chlorpyrifos. For instance, Zhao et al. (2023) illustrated the effectiveness of microbial consortia in simultaneously degrading multiple pesticides, presenting a sustainable solution for pesticide cleanup. These studies highlight the potential of microbial bioremediation as a viable strategy to address pesticide contamination and promote environmental sustainability.

Groundwater Remediation with Bioaugmentation: Bioaugmentation involves introducing specific microbial strains into contaminated environments to enhance bioremediation processes. Recent advancements in molecular biology and genetic engineering have facilitated the development of tailored microbial consortia aimed at targeted pollutant degradation (Borchert et al., 2021; Chaudhary et al., 2023). Chettri and Verma (2024) reviewed various microbes and plant-based bioremediation applications, highlighting the significant role of bioaugmentation in remediating groundwater contaminated with chlorinated solvents, hydrocarbons, and other industrial pollutants. A notable example of bioaugmentation's success is the remediation of trichloroethylene (TCE) plume using these techniques, documented by Ellis et al. (2000) and reviewed in recent advancements by Wu et al. (2022). These studies underscore the effectiveness of bioaugmentation strategies in enhancing bioremediation outcomes and advancing environmental cleanup efforts.

Nanotech Boost: Nanotechnology has emerged as a pivotal tool in enhancing bioremediation processes by augmenting the degradation of pollutants through innovative approaches. Studies by Mandeep and Shukla (2020) and Patel et al. (2022) underscore the transformative impact of nanoparticles in this field. Nanoparticles play multifaceted roles in bioremediation. They can serve as carriers, facilitating the delivery of nutrients or electron acceptors to microbial communities, thereby enhancing their metabolic activities and optimizing pollutant degradation efficiency (Thangavelu & Veeraragavan, 2022). Moreover, nanoparticles possess intrinsic properties that enable them to interact directly with contaminants, thereby increasing their bioavailability and susceptibility to enzymatic breakdown by microorganisms (Hidangmayum et al., 2023).

The integration of nanotechnology with bioremediation strategies represents a sophisticated approach to addressing environmental contamination challenges. By leveraging the unique characteristics of nanoparticles, such as their high surface area-to-volume ratio and tunable surface chemistry, researchers can design tailored solutions to improve remediation efficacy across diverse environmental matrices. This advancement holds promise for enhancing sustainability in environmental management practices and fostering cleaner ecosystems.

Targeting “Forever Chemicals”: Addressing “Forever Chemicals”: Jain et al. (2023) in their editorial delve into bioremediation strategies

tailored for synthetic organic compounds (SOCs), particularly focusing on challenging-to-degrade man-made chemicals such as PFAS (per- and polyfluoroalkyl substances). Ongoing research aims to identify the engineered microbes capable of effectively breaking down these persistent pollutants (Li, J., Li, X et al., 2022; LaFond, et al., 2023).

The editorial underscores the urgency and complexity of remediation efforts against SOCs, emphasizing the need for innovative biotechnological approaches to mitigate environmental contamination. Studies like those referenced highlight advancements in understanding microbial capabilities and genetic engineering techniques, paving the way for targeted and sustainable solutions in environmental cleanup practices.

Types of bioremediations

Bioremediation can be broadly classified into 2 types based on where remediation is carried out (Fig. 1).

In situ bio remediation

In-situ bioremediation stands as a crucial method for cleansing hazardous compounds from the environment (De et al., 2014). When remediation occurs directly at the source of pollution, it falls under the category of in-situ remediation. If this remediation employs flora and fauna, it is referred to as bioremediation. On-site bio-processing, termed in situ bio-processing, addresses contaminated soil or groundwater without the need for extensive and costly cleanup efforts. This approach harnesses harmless microorganisms, plants, and animals to remove chemical toxins effectively. Opting for local bioremediation offers notable advantages, including reduced physical labour, minimized haulage expenses, and the utilization of natural processes.

Several options for on-site bioremediation exist, including natural attenuation, composting, slurping, bioventing, sparking, and bacteria that aid in phytoremediation, among others. These methods align with one or more of the three fundamental in-situ bioremediation methodologies: bio-attenuation, bio-stimulation, or bio-augmentation (Maitra, 2018). Despite its benefits, the efficacy of in-situ bioremediation can be limited by various factors. Soil contamination with diverse organic compounds is common, particularly in industrialized areas. Furthermore, organic compound degradation can be hindered by factors such as low temperatures, anaerobic environments, nutrient and co-substrate deficiencies, a lack of degradation potential, and the bioavailability of organic compounds (De et al., 2014).

Environmental considerations significantly influence the success of in-situ bioremediation efforts. Factors such as soil type, neutral pH, seasonal temperature variations, nutritional supplements, oxygen levels, and gene expression play pivotal roles (Sharma, 2019; Romantschuk et al., 2023). By addressing these factors effectively, in-situ bioremediation can emerge as a powerful tool for mitigating environmental contamination and promoting ecosystem restoration.

In-situ bioremediation techniques

In-situ bioremediation entails the extraction of toxins through the action of microorganisms, which facilitate their removal and biotransformation. Over recent decades, numerous soil remediation technologies have emerged, including conventional physical and chemical methods known for their efficiency and cost-effectiveness. However, biological treatments such as phytoremediation, bioremediation, and term remediation have gained prominence due to their cost-effectiveness, environmental friendliness, and relatively shorter timeframes. Despite the attention garnered by these natural processes and their outcomes, there remains limited understanding of the interplay among various soil purification techniques (Urionabarrenetxea et al., 2021) as illustrated in Fig. 2.

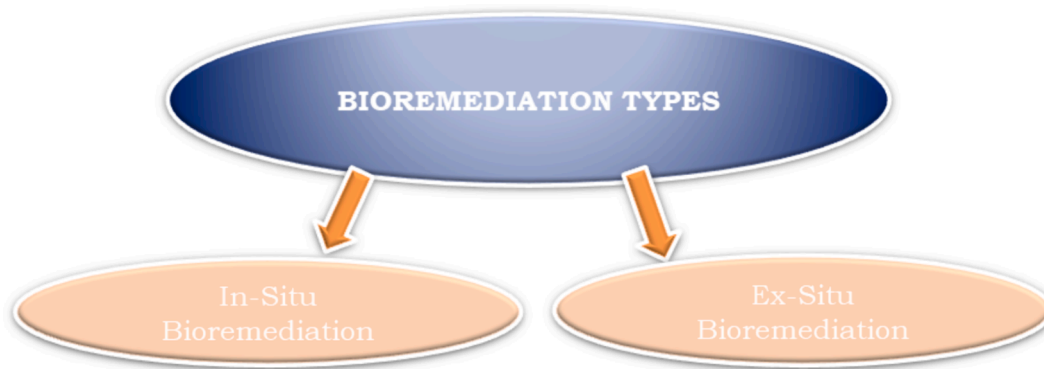


Fig. 1. Bio-remediation and its types.

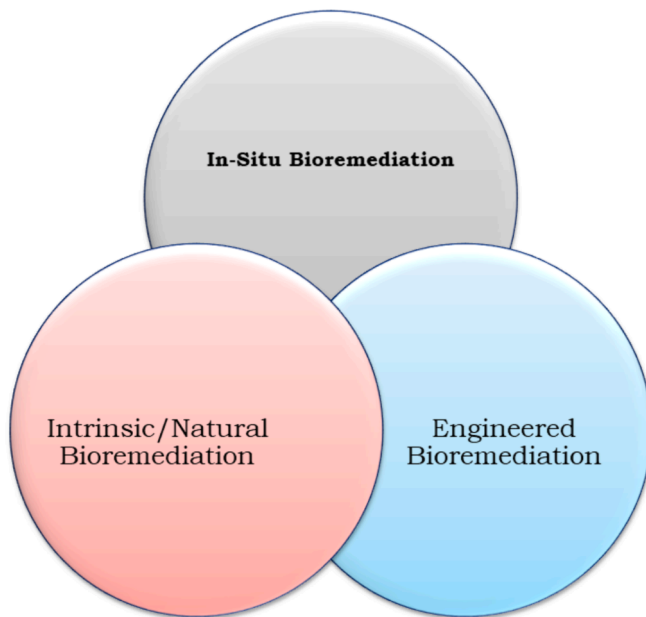


Fig. 2. Classification of In-situ conservation.

Methods of in-situ bioremediation

Intrinsic In-Situ bioremediation

Intrinsic bioremediation harnesses the natural capability of indigenous bacteria to metabolize toxins without external manipulation. This process involves converting environmental pollutants into harmless forms by leveraging the inherent capacity of existing microbial populations. Particularly effective in soil environments plagued by continuous contamination from pollutants and chemicals, intrinsic bioremediation is also utilized in subsurface locations such as underground fuel tanks. However, prior to implementing intrinsic bioremediation, it is crucial to verify and assess the microbial community's inherent ability to degrade pollutants through in-vitro testing.

Intrinsic/Natural bioremediation techniques

Bioremediation is a natural method that harnesses the inherent capabilities of flora and fauna to mitigate toxins present in contaminated lithosphere, hydrosphere, and atmosphere without any external scientific manipulations. Certain organisms intrinsically demonstrate bioremediation, rendering this method highly cost-efficient, environmentally friendly, and reliable. Continuous monitoring, analysis, and the

introduction of nutrients, native or exogenous microbes, as well as consideration of geochemical and physical soil features, are essential aspects of this process. Different types of natural or Intrinsic bioremediation are given in Fig. 3. Isolating indigenous bacteria and evaluating physical characteristics are crucial steps for predicting successful site mitigation. Strategies such as bioventing can facilitate the bioremediation process, with aerobic processes serving as an effective approach for in-situ bioremediation. These processes provide oxygen to the soil through various means, including composite sparging, among others, as depicted in Fig. 3 (Donlon and Bauder, 2006; Kaniserry and Sims, 2011).

Types of intrinsic / natural bioremediation techniques

Phytoremediation

Phytoremediation technologies employ living plants to remove impurities from contaminated soil, air, and water (Ramya et al., 2023). These techniques provide effective solutions for remediating heavy metal-polluted soil, including phytostabilization, where plants reclaim heavy metals in the soil, and phytoextraction, which involves using plants to extract and remove heavy metals from the soil. Various phytoremediation methods, such as phytodegradation, phytovolatilization, phytoextraction, and phytotransformation, are widely used and discussed within the context, as illustrated in Fig. 4 (Hassan et al., 2024).

Advantages and limitations of phytoremediation

Phytoremediation, renowned for its cost-effectiveness and economic

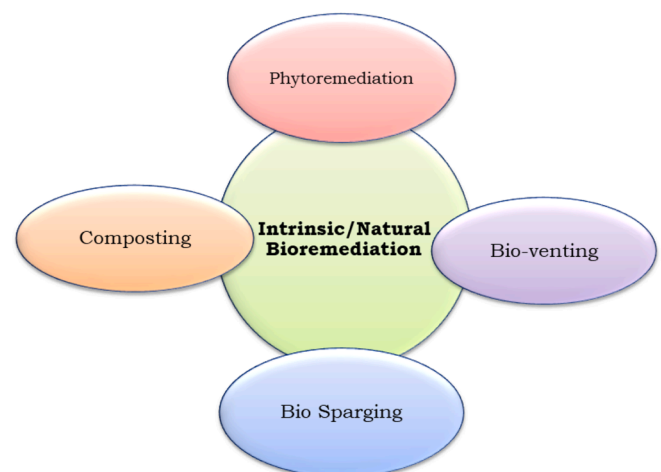


Fig. 3. Intrinsic /Natural bio-remediation and its division.

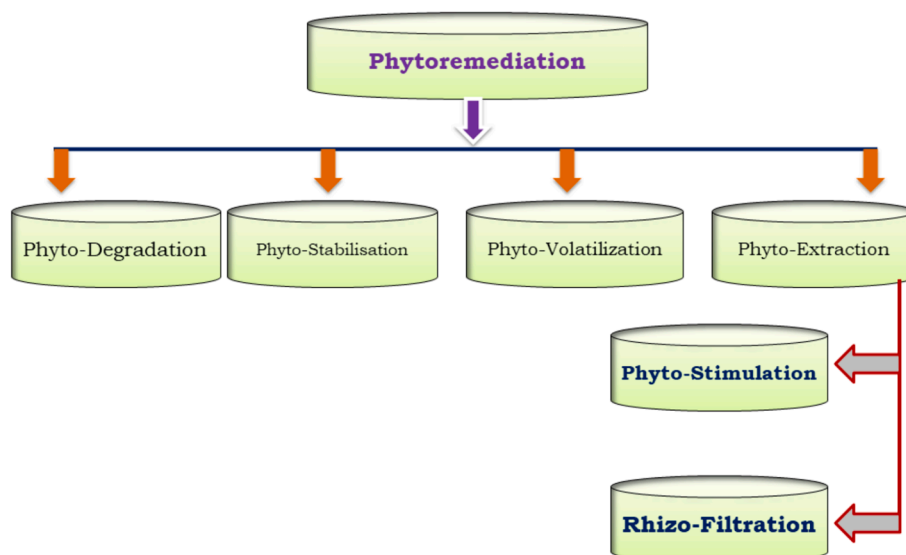


Fig. 4. Different types of phytoremediation.

viability compared to alternative techniques, presents numerous advantages in environmental restoration endeavors. Firstly, it enables phytomining companies to recover and recycle valuable metals, thereby contributing to both environmental remediation and resource conservation. Additionally, phytoremediation plays a pivotal role in preserving the fertility of topsoil, enriching plant phytochemicals, and enhancing soil health. By mitigating soil leaching and erosion, this approach safeguards the integrity of the surrounding ecosystem.

However, despite its merits, phytoremediation has limitations. One significant constraint is its primary suitability for surface areas, limiting its effectiveness in deeply contaminated soils. Additionally, the presence of toxic substances, particularly metals, in contaminated soil can adversely affect the plants used in phytoremediation, potentially compromising the quality of plant-derived products such as food and cosmetics. Therefore, careful consideration is necessary during the disposal of affected plants to minimize further contamination or environmental hazards associated with their disposal. In summary, while phytoremediation shows promise for environmental restoration, addressing its limitations and implementing appropriate management practices are crucial to ensure its effectiveness and mitigate potential risks (Madhav et al., 2024).

Phytodegradation

Phytodegradation utilizes plants' capabilities to remediate soil contaminated with heavy metals and toxins by breaking down organic pollutants into less harmful forms. This process involves the enzymatic breakdown of organic compounds by plant roots, followed by their absorption into the plant. Consequently, contaminated and barren soil is converted into fertile, cultivable soil, a phenomenon often termed phytotransformation. Transpiration, a crucial physiological process in plants, facilitates the absorption and upward movement of sap, thereby aiding in contaminant uptake. The rate of transpiration is influenced by various ecological factors such as temperature, pH, nutrient availability, soil moisture, and others, underscoring the intricate interplay between plant physiology and environmental conditions in phytodegradation processes (Van Thuan et al., 2023; Pandey et al., 2024).

Phytostabilization

Phytostabilization involves metal-tolerant plant species binding heavy metals to their roots through processes like precipitation, absorption, and sequestration within root tissues. This method promotes

plant growth and contributes to restoring soil health, even in environments contaminated with heavy metals. A significant advantage of phytostabilization is its capability to reduce the necessity for biomass disposal, as plants can convert harmful chemical components into less toxic forms through chemical reactions and the release of chemical agents. Additionally, phytostabilization can be strengthened by incorporating organic and inorganic soil amendments (Borbón-Palomares et al., 2024). These amendments alter the soil's pH and physicochemical properties, improving water holding capacity and facilitating plant colonization. Moreover, the presence of microorganisms such as certain bacteria and mycorrhizae in the roots enhances the efficiency of phytoremediation processes.

Phytovolatilization

Phytovolatilization is a technique used to detoxify toxins in the soil by converting them into gaseous compounds that are then emitted into the atmosphere. Plants from the Brassicaceae family are particularly effective in this process, making them valuable for volatilizing pollutants from the soil. It's important to note that while phytovolatilization does not completely eliminate toxins from the soil, it effectively transfers pollutants from the soil to the atmosphere (Sharma et al., 2024).

The process of phytovolatilization begins with the absorption of heavy metals by plant roots during transpiration. Once absorbed, these metals undergo transformation into ionic forms within the plant. Subsequently, these transformed compounds are released into the atmosphere as volatile gases. This mechanism allows for the gradual removal of toxins from the soil, reducing their concentration and potential environmental impact (Ionata et al., 2024).

Phytoextraction

Phytoextraction, a prominent bioremediation technique, aims to remove contaminants from the soil, particularly at or above ground level. Also known as phytoaccumulation and phytosequestration, phytoextraction can occur naturally or be induced by chemical agents that enhance the availability of metals in the soil. The accumulation of metals in plants depends on the species, with the rate of phytoextraction influenced by factors such as plant selection, metal availability in the soil, and plant performance (Thalassinios et al., 2024).

Phytoextraction methods involve several key processes, including mobilization of the rhizosphere, uptake of heavy metals by plant roots, translocation of metals within the plant, and compartmentalization of

contaminants. These processes collectively contribute to effectively removing contaminants from the soil, thereby supporting soil remediation efforts (Medina-Díaz et al., 2024).

Phytostimulation

Phytostimulation involves enhancing the microbial flora within the soil to boost their ability to effectively degrade contaminants. This process primarily occurs in the rhizosphere, the area of soil influenced by plant roots. Plants release biomolecules such as carbohydrates and acids into the rhizosphere, which stimulate the activity of microorganisms responsible for breaking down pollutants (Ratajczak et al., 2023).

These biomolecules act as potent stimulants, promoting the proliferation and metabolic activity of soil microbes. This increased microbial activity accelerates the degradation of contaminants in the soil. Phytostimulation has shown significant effectiveness in remediating hydrocarbon pollutants by enhancing the degradation capabilities of microbial communities in contaminated environments. Overall, phytostimulation represents a promising strategy for augmenting natural remediation processes in soil ecosystems (Gill et al., 2023).

Rhizofiltration

Rhizofiltration is a phytoremediation technique that utilizes a dense network of plant roots to filter water, removing undesired compounds and nutrients. Contaminants are either absorbed or adsorbed by the roots and retained indefinitely. The process begins by cultivating rhizofiltration plants hydroponically in clean water to establish a robust root system. Once the roots are well-developed, the plants are transitioned to contaminated water for acclimatization before returning to clean water (Singh et al., 2024).

Following acclimatization, the plants are transferred to contaminated sites where they extract heavy metals and other pollutants from the soil or water. The roots of these plants accumulate heavy metals, and afterward, the plants are harvested and disposed of using appropriate techniques to ensure safe handling of the contaminants. Rhizofiltration is particularly effective for purifying contaminated groundwater, either through direct planting at the site or by transporting polluted water to a controlled rhizofiltration setup.

In all cases, plants are often cultivated under controlled conditions, such as in greenhouses, to optimize their effectiveness in removing pollutants. This controlled environment ensures that plants are well-prepared to handle specific contaminants and environmental conditions, thereby enhancing the overall efficiency of the remediation process (Tapia et al., 2024).

Bioventing

In the 1990 s, bioventing emerged as a significant technology for soil remediation and has since found widespread commercial application. Bioventing accelerates aerobic degradation by supplying oxygen to native microorganisms, enabling them to break down organic contaminants adsorbed in soil under oxygen-depleted conditions (Donlon and Bauder, 2006).

This method involves injecting air directly into the contaminated zone through vertical and horizontal wells, aiming to use just enough air to stimulate degradation while minimizing the volatilization and release of contaminants into the environment. Bioventing can introduce oxygen into the subsurface above the water table through wells drilled into contaminated soil, either by pulling air in or blowing it through these wells. Moreover, nitrogen and phosphorus may be added to enhance microbial growth rates, creating conditions favorable for bioremediation by transforming toxins into less hazardous forms (Hellekson, 1999).

Bioventing is commonly used to remediate aerobically degradable organic compounds like fuel residues in unsaturated soil zones. Its effectiveness correlates with soil permeability, although it diminishes in

diverse soils where contaminants may persist in less permeable layers. Competition from organic matter or iron oxidation in soil can further reduce its efficiency in soils rich in organic matter and iron content (Triozi et al., 2024). Compared to Soil Vapor Extraction (SVE), bioventing typically requires a longer remediation period, ranging from several years to decades, especially in homogeneous sandy soils.

Biosparging

This methodology is similar to bioventing in that it involves delivering air into the soil to enhance microbial activity and thus promote the removal of pollutants from contaminated areas. However, unlike bioventing, biosparging involves pumping air into the saturated zone of the soil. This process allows vapours of hydrocarbons to migrate upward into the unsaturated zone, where biodegradation is further promoted. The success of biosparging depends on two critical parameters: soil porosity, which affects the bioavailability and biodegradability of microbial pollutants. By supplying oxygen to the soil's microorganisms, biosparging accelerates the natural biodegradation of aerobic chemicals.

Air injection oxygenates the contaminated saturated soil zone, increasing oxygen concentration and promoting biodegradation. This process converts pollutants into non-toxic soil compounds, eliminating the need for air extraction and treatment (Azubuike et al. 2016). Biosparging utilizes low airflow rates to maintain a sufficient supply of oxygen for microbial activity. It is a well-established and widely used method for remediating aerobically degradable organic contaminants, such as fuel waste in saturated soils. The efficiency of biosparging increases with soil permeability; however, it is less effective in diverse soils where contaminants persist in less permeable layers.

In soil, organic oxidation competes with contaminant oxidation, making biosparging less efficient in soils with high organic matter and iron content. Introducing air into the soil enhances the microorganisms' ability to degrade contaminants, but their capacity to do so depends on the soil's permeability (Da Silva et al. 2020; Yaashikaa et al., 2024).

Composting

Composting is an effective bioremediation method where microorganisms convert organic substances and pollutants in soil into less harmful byproducts. Elevated temperatures during composting enhance the biodegradation process, speeding up the transformation of toxic compounds. This technology is applied in two primary ways to manage polluted soils or sediments: composting contaminated soils for efficient degradation and increasing the porosity of the compost mixture. To achieve this, soil is blended with materials such as wood, animal manure, vegetable waste, and specific organic matter, maintaining thermophilic temperatures above 50 °C to ensure the decomposition of organic materials. Techniques like aerated static heaps and mechanically agitated vessels are used to remediate individual "batches" of polluted soil or sediment, with windrow composting being the most cost-effective method (Noor et al., 2024).

Composting offers several benefits, including the sustainable use of biodegradable organic waste, which improves soil structure, mineral content, and microbial activity. However, it also has limitations, particularly regarding the detoxification of heavy metals, which is crucial for plants used in remediation. Plants respond to heavy metal toxicity through avoidance and tolerance mechanisms. Once absorbed, detoxification occurs in the cytosol using organic and inorganic compounds present, ensuring the plant's ability to thrive and contribute to soil remediation (Ottani et al., 2024).

Engineered bioremediation and its types

Engineered bioremediation and its types Fig. 5. Plays a significant role in bioremediation

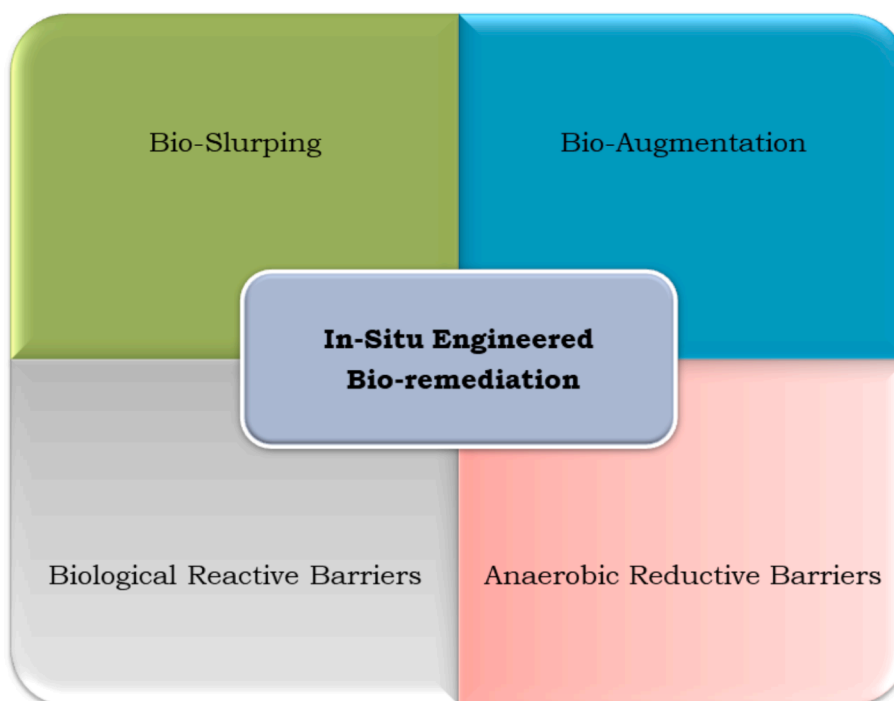


Fig. 5. In-situ Engineered Bio remediation and types.

Bio-slurping

This method integrates vacuum-assisted pumping, soil vapor extraction, and bioventing to indirectly treat soil and groundwater by delivering oxygen to enhance biodegradation. It targets the recovery of free-phase contaminants like light non-aqueous phase liquids (LNAPLs) across capillary, unsaturated, and saturated zones. Moreover, this technique effectively remediates soils contaminated with volatile and semi-volatile organic pollutants.

The process employs a “slurp” mechanism to extend the free product layer and extract fluids (free products and soil gases), akin to drawing liquid through a straw from a tank. The pumping mechanism raises LNAPLs to the surface for separation from air and water (Naseem et al., 2024). This system enables comprehensive remediation of accessible contaminants, similar to conventional bioventing systems. However, excessive soil moisture can reduce transpiration efficiency, decrease required oxygen levels, and lower microbial activity.

While this approach may not suit low-permeability soils due to limited reinfiltration potential, it offers cost savings in groundwater operations by minimizing storage, treatment, and disposal expenses. A significant challenge of this in-situ technology lies in establishing an effective treatment zone in deep, permeable areas with fluctuating water levels, hindering complete air saturation of the contaminated lens (Saravanan et al., 2024, Fig. 5).

Bio-Stimulation

The term “biostimulation” commonly refers to the addition of electron acceptors, electron donors, or nutrients to enhance the biodegradation activity of indigenous microbial populations in soil. The supplemented nutrients stimulate bacterial populations, increasing their ability to degrade macromolecules into essential micro and trace elements. This technique is an effective, cost-efficient, and environmentally friendly approach to remediation. Biostimulation involves the addition of highly concentrated nutrients such as nitrogen, phosphorus, oxygen, and electron donors to heavily contaminated areas to facilitate the breakdown of hazardous and toxic microorganisms (Goswami et al.,

2018; Muter, 2023).

Benefits and challenges of Biostimulation

Biostimulation offers significant advantages in bioremediation by leveraging pre-existing, well-suited indigenous microorganisms dispersed throughout the subsurface, resulting in highly efficient, cost-effective, and environmentally friendly remediation processes.

However, challenges exist with biostimulation, primarily concerning the distribution of additives to subsurface microorganisms. The effectiveness of nutrient dispersal is heavily reliant on the surface geology of the area, with dense and impermeable subsurface lithology posing challenges for achieving uniform distribution. Additionally, accurately quantifying the amount of nutrients required can be difficult, and the presence of cracks or gaps in the subsoil may create preferential pathways for additives, hindering their even dispersion (Adams et al., 2015).

Bio-Augmentation

Bioaugmentation involves introducing genetically modified or naturally occurring microbial strains into contaminated sites to enhance the degradation or elimination of undesirable microorganisms. Specific strains of microflora are cultured and added to facilitate contamination remediation. Unlike biostimulation, which involves supplying nutrients to support indigenous bacteria, bioaugmentation utilizes supplementary microbes, such as bacteria or archaea, to accelerate pollution breakdown. Mixed bacterial cultures are often preferred due to their enhanced potential for degradation and adaptability compared to pure bacterial strains. Therefore, bioaugmentation frequently serves as a bioremediation alternative in oil-contaminated environments (Omenna et al., 2024).

Factors influencing bio-augmentation

Soil bioaugmentation with a limited quantity of pollutant-degrading microorganisms and multifunctional chemicals proved to be highly effective. The success of the bioaugmentation process primarily hinges

on the microbial consortium's adaptation to the specific conditions of the decontaminated site (Adams et al., 2015), particularly within the microenvironment.

Advantages and limitations of In-Situ bioremediation

In-situ bioremediation emerges as a preferred approach for environmental cleanup endeavours due to several compelling advantages. Primarily, it boasts cost-effectiveness compared to ex-situ methods, as it mitigates expenses related to material transport and infrastructure development. Moreover, its capacity to manage substantial volumes of contaminants directly at the source streamlines logistical complexities associated with material transportation.

Additionally, in-situ bioremediation yields notable recycling benefits, generating byproducts that can be utilized on-site, thereby promoting resource conservation and sustainable practices. Nonetheless, this method is not without its limitations. Its reliance on natural processes results in a longer remediation timeline compared to other techniques.

Furthermore, in-situ bioremediation may encounter challenges stemming from seasonal variations in microbial activity, exacerbated by unpredictable environmental factors and specific contaminant characteristics. Consequently, the remediation process may be prolonged, especially when dealing with less readily degradable contaminants, potentially spanning several years. Despite these challenges, in-situ bioremediation remains a pivotal tool in environmental restoration efforts, offering a balance of cost-effectiveness and sustainability for contaminated sites.

Ex-Situ bioremediation

Ex-situ bioremediation techniques involve the excavation and removal of contaminants from polluted sites for treatment elsewhere. The selection of ex-situ bioremediation methods depends on various factors such as treatment costs, depth of contamination, type and level of pollutants, location, and topography of the contaminated area. Performance considerations also play a crucial role in determining the most

suitable ex-situ bioremediation approach (Atlas and Philp, 2005). These methods encompass a range of techniques tailored to specific pollutant types. Solid-phase soil management techniques include landfilling, soil biopiling, and composting. Soil slurry phase treatment, on the other hand, involves bioreactor processing of the slurry phase (Maitra, 2018). Ex-situ bioremediation is effective for a wide array of hydrocarbon pollutants, including aliphatic chains, lubricating oils, carbon compounds, ketones, phenols, cresols, polycyclic aromatic hydrocarbons, and semi-volatile organic compounds (SVOCs). It serves as a valuable tool for remediating contaminated sites affected by such pollutants.

Ex-Situ bioremediation techniques

Ex-situ biodegradation methods involve the extraction of contaminants from polluted areas and their transportation to another location for treatment. The selection of ex-situ biodegradation techniques relies on performance criteria, as outlined by Atlas and Philp (2005). Factors influencing the choice of ex-situ bioremediation methods include the type and concentration of contaminants, treatment costs, geographical location, and topography of the contaminated site (see Fig. 6).

Slurry phase bioremediation

Slurry phase bioremediation is notably faster than other remediation methods, making slurry bioreactors a crucial type of ex-situ technology. These bioreactors are particularly effective for treating recalcitrant soil contaminants under controlled environmental conditions, as evidenced by studies from Mueller et al. (1991), Christodoulatos and Koutsospyros (1998), and Mohan et al. (2006).

The process begins by analyzing the contaminant fractions, the extent of biodegradation, and the physical nature of the soil. This information is used to create a slurry by mixing water with the excavated soil, forming a sludge. The slurry typically contains 5–40 % solids by weight, depending on the specific requirements of the bioreactor. The soil is pre-washed to remove fine contaminants, which, along with the wash water, are treated within the slurry bioreactor.

The bioreactor is then supplemented with necessary nutrients,

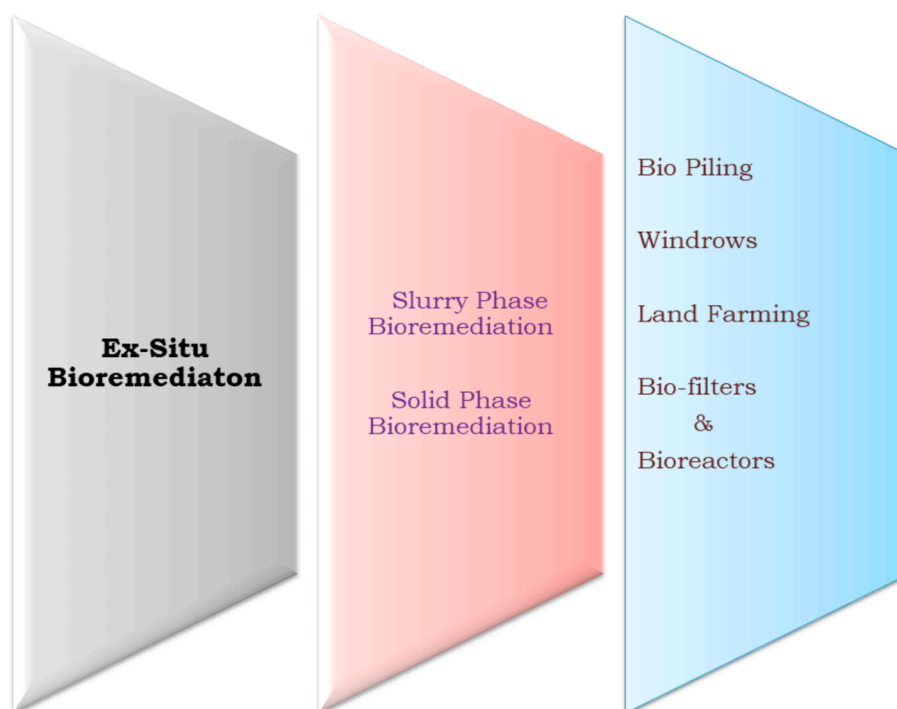


Fig. 6. Ex-situ Bio remediation and Its types.

oxygen, microorganisms, and pH adjusters (acid or alkali) based on the treatment conditions. Once biodegradation is complete, the slurry is dewatered, and the resulting liquids are filtered and clarified.

Slurry phase bioreactors have proven effective in bioremediating soils contaminated with pesticides. For instance, a study by Cookson (1995) demonstrated that, after two weeks of treatment, the concentration of pesticides such as 2,4-D, MCPA, alachlor, trifluralin, and carbofuran in contaminated soils could be reduced from 800 mg/kg to 20 mg/kg (Zhang et al., 2000).

Further, Nishino (1999) reported significant reductions in soil contaminants: from 14,715 mg/kg to 1,125 mg/kg for 2,4-dinitrotoluene (2,4-DNT) and from 8,940 mg/kg to 4,800 mg/kg for 2,6-dinitrotoluene (2,6-DNT) within two days. However, for soils contaminated with lower levels of 2,4-DNT (570 mg/kg) and 2,6-DNT (380 mg/kg), the reduction rate was about 40 %.

Overall, slurry phase bioreactors provide an effective and efficient means of bioremediating heavily contaminated soils, leveraging controlled conditions to accelerate the breakdown of complex pollutants.

Solid-Phase treatment

Solid-phase bioremediation, an ex-situ technology, involves excavating contaminated soil and forming it into heaps. These heaps are supplemented with organic waste such as leaves, animal dung, agricultural residues, and household, industrial, and municipal waste. Pipes are strategically placed throughout the heaps to facilitate air circulation, which is crucial for ventilation, microbial respiration, and growth.

Compared to slurry-phase processes, solid-phase systems require significantly more space and additional time to achieve complete remediation. Examples of solid-phase treatment methods include biopiles, windrows, land farming, and composting (Kulshreshtha et al., 2014). These methods leverage natural microbial processes to degrade contaminants, but their efficiency can be influenced by factors such as heap size, aeration, and the nature of the contaminants.

Biopile

Biopiling is an advanced technology that involves the accumulation of excavated contaminated soil, followed by the addition of nutrients and, in some cases, aeration to enhance bioremediation by promoting microbial activity. This method includes several key components: irrigation, aeration, nutrient addition, leachate collection systems, and treatment beds. Its popularity is growing due to its cost-effectiveness and efficiency in biodegradation when nutrients, temperature, and aeration are properly managed (Whelan et al., 2015).

The biopile method has been successfully applied to various soil types, including clay and sand (Chemlal et al., 2013; Akbari and Ghoshal, 2014). Enhancing the composition of the biopile can prevent resilience and boost microbial activity, thereby increasing the rate of biodegradation. Sanscartier (2009) observed that maintaining adequate humidity in a biopile reduces leaching and volatilization of less biodegradable contaminants, resulting in a lower final concentration of total petroleum hydrocarbons (TPH). Additionally, biopiles can handle large quantities of contaminated soil in a confined area (Chemlal et al., 2013).

Before processing, sieving and aerating the contaminated soil are critical for biopile performance (Delille et al., 2008). Bulking agents such as wood particles, hay, chips, bark, and other chemical-free components are often used (Rodríguez et al., 2010). Despite the advantages of biopiles over other ex-situ bioremediation methods, challenges such as complex engineering, maintenance, and operational costs remain distinctive to biopile systems.

Moreover, excessive heating of the air can lead to soil drying, which inhibits microbial activity and favors volatilization over biodegradation (Sanscartier et al., 2009). Although biopile systems save space compared to other ex-situ strategies like land farming, they are limited by energy

supply issues, especially at remote sites, which can hinder the uniform distribution of air through the pumps into the contaminated soil. Excessive soil heating further reduces microbial activity and increases volatilization rather than biodegradation.

Windrows

Windrows are a bioremediation technique that involves the periodic turning of heaped soil to enhance the microbial degradation of hydrocarbons present in contaminated soils. Regularly turning the waste piles improves aeration, promoting microbial growth and thereby accelerating decomposition. The pace of bioremediation can be increased by increasing aeration, adding water, ensuring a consistent supply of nutrients and pollutants, and incorporating periodic rotations to acclimate, convert, and mineralize the contaminants. This method has demonstrated a high rate of hydrocarbon removal (Vickers, 2017).

However, the windrow technique is less effective for soils contaminated with volatile compounds. Periodic turning can create anaerobic zones within the stacked soil, leading to the release of greenhouse gases such as methane (CH₄) and limiting effective air ventilation (Hobson et al., 2005).

Land farming

Land farming is a straightforward method of bioremediation commonly employed in agricultural settings. Typically conducted as ex-situ bioremediation, it may occasionally involve in-situ approaches. The viability of land farming depends largely on the depth of contamination, as it is relatively cost-effective and requires minimal equipment. In ex-situ land farming, indigenous microorganisms facilitate the aerobic biodegradation of toxic soil contaminants on the soil surface (Silva-Castro et al., 2012).

This method is commonly used to remediate areas contaminated with hydrocarbons, particularly polyaromatic hydrocarbons. Biodegradation and volatilization are the primary processes involved in contaminant degradation (Da Silva et al., 2012; Cerqueira et al., 2014). An advantage of ex-situ bioremediation techniques like land farming is that they do not require extensive pre-cleansing assessments of the contaminated area, thereby shortening the initial step and reducing both labor and costs.

However, despite its simplicity, land farming also presents several drawbacks. These include the need for a large operating area, reduced microbial activity due to unfavorable environmental conditions, increased excavation costs, and limited efficiency in removing inorganic pollutants (Cerqueira et al., 2014; Maila and Cloete, 2004).

Biofilters

Biofilters have demonstrated effectiveness in treating wastewater and reducing air pollution, making them valuable tools in environmental bioremediation. These filters represent one of the earliest technologies developed for this purpose, employing a bed of media where pollutants are degraded by microorganisms (Schmidt and Anderson, 2017; Lebrero et al., 2014; Rabbani et al., 2016). Various materials such as manure, organic matter, tree twigs, gritty dirt, stones, or synthetic fibers can be used as bed materials.

One specific type of biofilter, known as a trickling filter, is commonly utilized for treating wastewater and other liquid effluents. This filter typically consists of a spherical, vertical tank made of aggregate, ceramic, or plastic materials. Within the tank, a central vertical tubing connects rotatably to spray nozzles located at the top of the tank, allowing for the distribution of wastewater (Schmidt and Anderson, 2017). Microorganisms within the biofilms formed on the surface of the filter media degrade contaminants present in the wastewater.

Studies have demonstrated the efficacy of trickling biofilters in removing substantial quantities of pollutants from contaminated air,

particularly focusing on the removal of 1-butyl alcohol. Industrial off-gas treatment using bio-trickling filters has shown promising results, achieving removal efficiencies of over 80 % for 0.4–1.2 g/m³(–) butanol levels (Schmidt and Anderson, 2017).

Bioreactors

A bioreactor functions as a vessel where organic substances undergo a series of biological processes, culminating in the production of specific byproducts (Akshaya et al., 2023). These reactors are employed in various operational modes such as batch, fed-batch, sequencing batch, continuous, and multi-stage processes. The controlled environment within a bioreactor provides ideal conditions for fostering bioremediation growth (Fig. 7).

In the context of remediating contaminated soil, bioreactors offer several advantages. These include precise control over pH levels, regulated temperature, reduced turbulence, enhanced aeration, and accurate management of substrate and inoculum concentrations. The adaptable design of bioreactors ensures optimal biological degradation while minimizing non-biological losses (Venkata Mohan et al., 2004).

Advantages of Ex-Situ bioremediation

Ex-situ bioremediation presents several advantages over in-situ techniques, rendering it a versatile and highly effective approach for environmental cleanup endeavors. One significant advantage lies in its capability to address a wider spectrum of pollutants and soil compositions. This adaptability enables ex-situ methods to manage diverse contamination scenarios with greater efficacy compared to in-situ approaches. Additionally, regulatory compliance and resource management are more straightforward with ex-situ bioremediation. The presence of dedicated infrastructure for remediation ensures unhindered access to resources, facilitating efficient and thorough cleanup operations.

The suitability of ex-situ bioremediation can often be readily assessed based on on-site evaluation data, streamlining decision-making processes and enabling prompt implementation of remedial measures. Furthermore, ex-situ bioremediation offers the flexibility to accommodate varying volumes of contaminated material. Whether contending with small-scale or large-scale contamination, the process can be customized to handle different volumes adeptly. Lastly, the ability to homogenize contaminated material in ex-situ.

Disadvantages of Ex-Situ bioremediation

Despite its advantages, ex-situ bioremediation presents certain drawbacks that warrant consideration when choosing remediation strategies. One significant disadvantage is the financial burden associated with soil excavation and transportation. Ex-situ techniques entail the excavation of contaminated soil and its transportation to treatment facilities, resulting in heightened costs and necessitating additional engineering efforts, thereby adding complexity to the remediation process.

Moreover, the processes of soil excavation and transportation in ex-situ bioremediation elevate the risks of material handling and potential worker exposure to contaminants. To effectively mitigate these risks, proper safety protocols and measures must be implemented. While ex-situ bioremediation offers numerous benefits, such as versatility and regulatory compliance, its drawbacks, notably increased costs and safety concerns, underscore the importance of meticulous planning and execution to ensure successful outcomes in environmental cleanup efforts (Table 2) contains comprehensive comparative summary of Insitu and Exsitu bioremediation.

Internet of Things (IoT), Artificial Intelligence, and biosensors in bioremediation

The integration of the Internet of Things (IoT), Artificial Intelligence (AI), and biosensors in bioremediation and waste management represents a transformative approach to environmental sustainability. Recent advancements in these technologies enable more efficient, precise, and effective processes. Here's a detailed exploration based on current research and future perspectives:

Internet of Things (IoT) in bioremediation and waste management

Current advancements in IoT include the deployment of devices such as sensors and smart meters to continuously monitor environmental parameters like pH, temperature, and pollutant levels. These devices transmit real-time data (Midhila ., et al., 2023), enabling immediate action and decision-making. IoT-enabled systems can also automate the control of bioremediation processes by regulating nutrient supply to microorganisms based on real-time data, thus optimizing conditions for pollutant degradation. Additionally, IoT facilitates remote monitoring and control of bioremediation sites, reducing the need for human intervention and allowing for better management of multiple sites

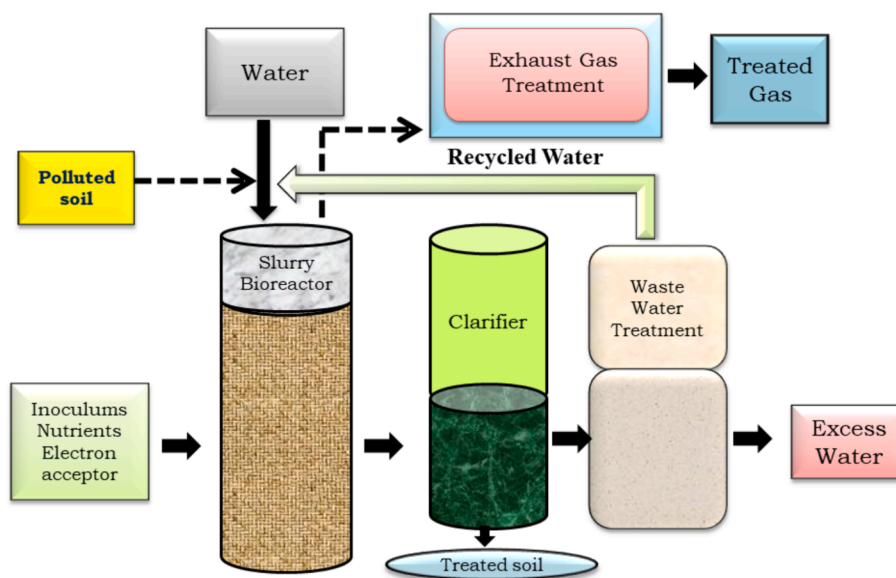


Fig. 7. Different types of bioreactor for various types of bioremediation.

Table 2
Comparison of various methods of insitu and exsitu bioremediation.

Sl no	Bioremediation Method	Principle/ Process	Advantages	Disadvantages	Area of application
In situ Bioremediation					
Types of Intrinsic / Naturalbioremediation Techniques					
1	Phytoremediation	Living plants remove impurities from contaminated soil, air, and water	Preserve soil fertility	Uptake of toxins by plants, Disposal of toxic plants	Heavy metal removal
2	Phytodegradation	Living plants break down organic pollutants into less harmful forms	Cost effectiveness, Recover valuable reuse Metals	Less efficacy in deeply contaminated soils, presence of toxins in plants	Environmental Restoration
3	Phytovolatalization	Transform and detoxify toxins to gaseous compounds	Conversion of toxins into volatile gases	Time consuming, potential harm to environment	Relocation of toxins from soil to atmosphere
4	Phytostabilization	Bind heavy metals through precipitation, absorption and sequestration	Soil health restoration	Final destination of contaminants	Biomass mitigation and disposal
5	Phytoextraction	Induced by chemical agents, plants extract toxins	Translocation and compartmentalization of heavy metals	The accumulation of metals is influenced by factors such as plant species and availability of heavy metals in the soil	Removal of contaminants above the ground level
6	Phytostimulation	Rhizosphere accelerated microbial degradation of toxic compounds by plants	Remediation of hydrocarbon pollutants	Steric inaccessability	Enhances Natural Bioremediation
7	Rhizofiltration	Absorption/Adsorption of toxic compounds by plant roots	Purification of contaminated ground water	Disposal of toxic roots	Controlled environments such as green houses
8	Bioventing	Injection or extraction of air into the soil	Create modified atmosphere to enhance Bioremediation	Longer duration ranging from a few years to several decades, particularly in homogeneous sandy soils.	Large scale bioremediation
9	Biosparging	Pumping of air into saturated zone of soil	Enhances the ability of micro organisms to degrade toxins	The success of biosparging depends on two critical parameters: soil porosity, which affects the bioavailability and biodegradability of microbial pollutants	Degradation of fuel waste in saturated soil
10	Composting	Microorganisms convert contaminants in the soil into less harmful byproducts.	Sustainable use of biodegradable organic waste, which enhances soil structure, mineral content, and microbial activity.	Maintenance of temperature	Management of polluted soils and sediments
11	Bioslurping	Integrated approach utilizing oxygen assisted vacuum-assisted pumping, soil vapor extraction, and bioventing for the indirect treatment of soil and groundwater	Extraction of fluids	Requirement of an effective treatment zone in deep, permeable areas with fluctuating water	Recovery of contaminants, such as light non-aqueous phase liquids, in capillary, unsaturated, and saturated zones.
12	Biostimulation	Addition of electron acceptors, electron donors, or nutrients to enhance the biodegradation activity of indigenous microbial populations in soil	Most effective, cost efficient and Environment friendly	The process depends upon the surface geology of the area	Heavily contaminated areas
13	Bioaugmentation	Use of genetically modified or natural microbial strains to enhance the degradation or elimination of undesirable microorganisms.	High efficiency with multifunctional chemicals in association with microorganisms	The process depends on microbial adaptation to the specific conditions of the decontaminated site	Oil contaminated Environments
Ex situ Bioremediation					
1	Slurry phase Bioremediation	Use of controlled conditions to accelerate the breakdown of complex contaminants.	Used for treatment of heavily contaminated soils	Maintenance of controlled conditions	Pesticide contaminated soils
2	Biopile	Advanced technology that involves the accumulation of excavated soil, followed by the addition of nutrients and, in some cases, aeration to enhance bioremediation by promoting microbial activity.	Comprehensive approach utilizing irrigation, aeration, nutrient addition, leachate collection systems, and treatment beds.	Limited by factors such as aeration, humidity and efficiency of biopile system	Used for treatment of specific soil types such and clay and soil
3	Windrows	Periodic turning of heaped soil to enhance the microbial degradation of hydrocarbons present in contaminated soils.	High rate of Hydrocarbon removal	Periodic turning creates anaerobic zones, release of green house gases such as methane	Removal of contaminants from natural environments
4	Land farming	Use of aerobic microbes to remove soil contaminants	Cost effective, requires minimal equipments	Large operating area, environmental conditions	Removal of inorganic pollutants from soil surface
5	Biofilters	Employment of media bed and removal of toxins through microbial conversion	Less space, more efficient, biofilm induced decomposition	Filter impairment, operation requires skilled persons	Waste water treatment
6	Bioreactors	Organic biological conversion, ultimately resulting in the production of specific products	Batch, fed batch and continuous processes may be operated	Maintenance of controlled conditions inside the bioreactor	Versatile and highly effective approach for environmental cleanup endeavors.

simultaneously. Looking ahead, the development of more sophisticated IoT networks could enhance data sharing and coordination among various stakeholders involved in waste management. IoT systems can also predict equipment failures or operational issues in bioremediation facilities, ensuring timely maintenance and reducing downtime.

Artificial Intelligence (AI) in bioremediation and waste management

AI contributes significantly by analyzing vast amounts of data collected by IoT devices, identifying patterns and correlations that human analysis might miss. This leads to a better understanding of contamination sources and more effective remediation strategies. AI models can predict the outcomes of different bioremediation approaches, helping to select the most efficient and cost-effective methods by considering variables like pollutant type, environmental conditions, and microbial activity. Furthermore, AI optimizes bioremediation processes by dynamically adjusting parameters such as oxygen levels, nutrient supply, and microbial inoculation based on real-time data. Future advancements in AI, including advanced machine learning techniques, are expected to create adaptive bioremediation systems that learn from ongoing processes and improve over time, increasing efficiency and effectiveness. Integration of AI with robotics could also lead to autonomous bioremediation units capable of conducting complex tasks with minimal human intervention.

Biosensors in bioremediation and waste management

Biosensors play a crucial role in detecting and measuring pollutants with high sensitivity and specificity. These sensors often use biological recognition elements like enzymes, antibodies, or nucleic acids. Biosensors can monitor the activity and viability of microorganisms used in bioremediation, ensuring they function optimally to degrade pollutants. Additionally, biosensors continuously track environmental conditions, providing data that can be used to adjust bioremediation strategies in real time. Future prospects include the development of integrated sensor networks combining multiple biosensors to provide comprehensive monitoring of various pollutants and environmental parameters. Advancements in wearable and portable biosensors will enable field personnel to conduct on-site monitoring and analysis, enhancing the responsiveness and flexibility of bioremediation efforts.

Integration and synergy

The synergistic integration of IoT, AI, and biosensors can revolutionize bioremediation and waste management. Combining IoT for data collection, AI for data analysis and decision-making, and biosensors for precise monitoring can create smart bioremediation systems that are highly efficient and adaptive. These technologies can establish real-time feedback loops where data from biosensors inform AI models, which then adjust IoT-controlled systems to optimize bioremediation processes continuously. Such integrated systems are not only more sustainable by reducing waste and resource usage but also scalable, making them applicable to a wide range of bioremediation and waste management scenarios globally.

Projects that have effectively deployed AI and the Internet of Things (IoT) in pilot batches

Project BioSCOPE in the European Union (EU) aims to develop an integrated bioremediation platform utilizing IoT, AI, and biosensors to remediate contaminated soil and groundwater across multiple sites. The project employs IoT sensors to monitor environmental parameters like pH, temperature, moisture, and contaminant levels, while AI analytics leverage machine learning algorithms to analyze real-time sensor data and optimize bioremediation processes. Integrated biosensors detect microbial activity, assessing the effectiveness of bioremediation

treatments. BioSCOPE has demonstrated significant improvements in bioremediation efficiency and cost-effectiveness compared to traditional methods. The integrated platform provides real-time insights, enabling adaptive management of remediation efforts and achieving successful contaminant reduction at several pilot sites across the EU (refer [Table 3](#)) for detailed projects.

The **Smart Bioremediation System** project, located at an industrial site in the United States, aims to develop an advanced bioremediation system using IoT, AI, and biosensors to treat petroleum hydrocarbon-contaminated soil. This innovative system features IoT-enabled sensors installed in the contaminated soil to monitor key environmental parameters such as soil moisture, temperature, oxygen levels, and contaminant concentrations. An AI-driven control system employs machine learning algorithms to analyse sensor data and autonomously adjust nutrient dosing, aeration rates, and watering schedules to optimize microbial activity and contaminant degradation. Integrated biosensors monitor microbial populations and metabolic activity, providing real-time feedback on bioremediation effectiveness. The project demonstrated that the Smart Bioremediation System significantly outperformed conventional remediation methods, achieving rapid contaminant degradation and successful site remediation within a shorter timeframe by continuously optimizing environmental conditions based on AI-driven insights (refer [Table 3](#) for detailed projects).

The REBECA initiative, located at a contaminated aquifer site in Spain, aims to develop a remote monitoring and control system using IoT, AI, and biosensors to enhance the bioremediation of chlorinated solvent-contaminated groundwater. Key features include IoT-enabled groundwater monitoring through sensors deployed in monitoring wells to measure groundwater flow rates, contaminant concentrations, and geochemical parameters. An AI-based decision support system utilizes algorithms to analyze sensor data, predict contaminant plume behavior, and optimize the design and operation of in-situ bioremediation treatments. Integrated biosensors monitor microbial activity and redox conditions in the groundwater, providing insights into the effectiveness of bioremediation strategies. The REBECA initiative successfully demonstrated the feasibility and effectiveness of this integrated system for remote monitoring and control of bioremediation processes, leading to significant reductions in chlorinated solvent concentrations and the restoration of groundwater quality at the pilot site (for detailed projects refer [Table 3](#)).

These examples highlight successful pilot projects and initiatives that have leveraged IoT, AI, and biosensors to enhance bioremediation processes. By integrating real-time monitoring, data analytics, and adaptive control systems, these technologies have significantly improved the efficiency, effectiveness, and cost-effectiveness of environmental remediation efforts. This integration paves the way for more sustainable and innovative approaches to bioremediation.

Conclusion

This review underscores the significant advancements and transformative potential of bioremediation, positioning it as a sustainable and effective technology for pollution mitigation both now and in the future. By blending traditional bioremediation insights with modern technological innovations, the analysis highlights the multifaceted capabilities of bioremediation in tackling complex environmental challenges. Bioremediation leverages natural biological processes to decompose and detoxify hazardous substances, offering a cost-effective and eco-friendly alternative to conventional methods. The review examines various strategies, including in-situ and ex-situ techniques, and emphasizes phytoremediation, which utilizes plants for heavy metal removal, showcasing its broad applicability in treating industrial effluents, contaminated soils, and groundwater.

The integration of emerging technologies such as IoT, AI, and biosensors into bioremediation practices is thoroughly explored, demonstrating how IoT enables real-time monitoring and remote management,

Table 3
Comparative Analysis of Major Bioremediation Projects Worldwide.

Name of Bio-Remediation Project	Objective	Implications if not Implemented
Symbiorem	Remediation of pollutants: - Polycyclic aromatic hydrocarbons - Volatile aromatic hydrocarbons - Heavy metals - Mineral oil	- Neglecting this issue will lead to elevated occurrences of mental and physical illnesses. - It will also contribute to a rise in premature mortality rates. - Failing to take action will exacerbate public health challenges significantly.
Biosysmo	Bioremediation systems leveraging synergistic removal of mixed pollutants: - Use combinations of bacteria, fungi, and plants - Equipped with natural or engineered pathways for pollutant degradation - Employ a computationally-assisted analysis framework - For designing and optimizing synergistic biosystems	- Continued accumulation of pollutants in the environment poses significant risks to human health, leading to various mental and physical illnesses. - Persistent pollutants contribute to biodiversity decline, compromising ecosystems' ability to provide essential services such as carbon sequestration and contaminant removal. - Addressing pollutant accumulation is crucial to mitigate health risks and preserve ecosystem functions vital for sustainable environments.
Eiclar	An Integrated project - Integrates biological and non-biological processes - Improves sustainability and cost-effectiveness of in situ bioremediation (ISBR) for diverse contamination issues - Combines: - Physical processes - Electrical processes - Chemical processes - Enhances precision and value of environmental biotechnologies - Effectively addresses complex land contamination problems	- The sustainability and cost-effectiveness of in situ bioremediation (ISBR) for diverse contamination issues may stagnate. - Traditional methods may prove inadequate for addressing complex land contamination problems, prolonging environmental degradation and increasing remediation costs. - Insufficient precision in environmental biotechnologies could impede efforts to restore and sustain healthy ecosystems effectively.
Electra	Electricity-driven low energy and chemical input technology for accelerated bioremediation: - Develop and test advanced bio-electrochemical systems for remediation biotechnologies - Testing: Conducted at laboratory scale under environmentally relevant conditions - Deploy the four most efficient technologies in China and Europe	- Failure to implement the consortium's efforts would hinder the advancement of bio-electrochemical systems for remediation biotechnologies. - Slower progress in addressing environmental contamination issues in China and Europe could result from this missed opportunity. - Not deploying the most efficient technologies could prolong environmental risks and undermine sustainability efforts in both regions.
Greener	Assess and characterize polluted waters, soils, and sediments	The project's hindrance would restrict the comprehensive assessment and remediation of

Table 3 (continued)

Name of Bio-Remediation Project	Objective	Implications if not Implemented
	- Study microbial consortia for bioremediation purposes - Develop optimized bioremediation technologies - Implementation goals: - Demonstrate and scale hybrid bioremediation systems - Define business models for exploiting results - Ensure regulatory compliance - Maximize innovation impact for economic growth and job creation	- polluted waters, soils, and sediments. - Limitations in advancing bioremediation technologies and developing hybrid systems would impact environmental sustainability. - Delaying economic growth and job creation opportunities linked to innovative technologies could result from these setbacks.
Mibirem	- Standardize sample collection methods and enhance data management practices - Cultivate active degraders to optimize bioremediation processes - Publicly share genomic data to foster collaboration and transparency - Develop an IT modeling tool for informed bioremediation strategy formulation - Scale up microbiome analysis capabilities for comprehensive environmental assessments - Ensure regulatory compliance throughout project implementation - Establish market viability of the MIBIREM toolbox for widespread adoption in bioremediation	- Inconsistent sample collection and data management would hinder accurate bioremediation efforts. - Limited discovery of effective degraders and restricted scientific collaboration would slow advancements in the field. - Absence of an integrated IT modeling tool might reduce informed decision-making, prolonging environmental contamination issues and reducing overall efficacy in remediation efforts.

<https://eiclar.eu/about/>, 2024; <https://electraev.com>, 2024; <https://energy.ec.europa.eu>, 2024; <https://symbiorem.eu>, 2024; <https://www.biosysmo.eu>, 2024; <https://www.greener-h2020.eu>, 2024; <https://www.mibirem.eu>, 2024.

AI enhances predictive modeling and process optimization, and bio-sensors ensure precise pollutant detection and continuous environmental monitoring. The synergistic integration of these technologies creates intelligent bioremediation systems capable of real-time feedback and adaptive management, offering scalable solutions to pollution challenges and marking a significant stride towards sustainable environmental management. The review concludes by emphasizing bioremediation's pivotal role as an advanced, environmentally sustainable technology, combining traditional biological methods with cutting-edge advancements to address current pollution challenges and set the stage for future innovations. Ongoing research and development promise to enhance bioremediation's efficacy and applicability, making it a cornerstone of sustainable environmental practices with the potential to revolutionize pollution mitigation strategies and ensure a cleaner, healthier environment for future generations.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Chatgpt in order to restructure certain sentences for better articulation. After using this tool/service, the author(s) reviewed and edited the content as

needed and take(s) full responsibility for the content of the publication.

CRedit authorship contribution statement

Narendra Kuppan: Writing – original draft, Visualization, Validation, Conceptualization. **Midhila Padman:** Writing – review & editing, Validation. **Manjushree Mahadeva:** Visualization, Validation, Methodology, Data curation. **Subramani Srinivasan:** . **Raajasubramanian Devarajan:** Writing – review & editing, Visualization, Validation, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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